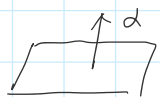


# Lesson 11 Root spaces

Wednesday, April 15, 2026 17:00



$$\alpha \in \Phi \quad (\Rightarrow (\alpha|\alpha) \neq 0)$$

Write  $V = \ker \alpha + F\alpha$

We can define reflection:  $S_\alpha: x \mapsto x - \frac{2(x|\alpha)}{(\alpha|\alpha)} \alpha$

From last time, we know that  $S_\alpha(A) = A$

Lemma  $\text{span}(A) = H^*$

pf: If  $\text{span}(A) \subsetneq H^*$ , then  $\exists 0 \neq h \in H$  s.t.  $\langle \text{span}(A), h \rangle = (0)$ .

i.e.  $h$  commutes with all  $L_\alpha$ 's. Since  $h$  commutes with  $H$ , this yields that  $h \in Z(L) = (0)$ , a contradiction.

Lemma  $\forall \alpha, \beta \in A, (\alpha|\beta) \in \mathbb{Q}$

pf: Recall that  $\frac{2(\alpha|\beta)}{(\alpha|\alpha)} \in \mathbb{Z}$ . we prove by show  $(\alpha|\alpha) \in \mathbb{Q} \quad \forall \alpha \in A$ .

$$\begin{aligned} \text{Indeed, } (\alpha|\alpha) &= (V^{-1}(\alpha) | V^{-1}(\alpha)) \\ &= \text{Tr}(\text{ad } V^{-1}(\alpha) \text{ ad } V^{-1}(\alpha)) \\ &= \sum_{\beta \in A} \beta (V^{-1}(\alpha))^2 \\ &= \sum_{\beta \in A} (\beta|\alpha)^2 \end{aligned}$$

$$\text{Then } \frac{1}{(\alpha|\alpha)} = \sum_{\beta \in A} \left( \frac{(\beta|\alpha)}{(\alpha|\alpha)} \right)^2 \in \mathbb{Q} \Rightarrow (\alpha|\alpha) \in \mathbb{Q}$$

Now consider  $\sum_{\alpha \in A} \mathbb{Q}\alpha \in H^*$

$\exists \alpha_1, \dots, \alpha_n \in A$  s.t.  $\{\alpha_1, \dots, \alpha_n\}$  is a basis of  $H^*$

For any  $\beta \in A$ , write  $\beta = \sum_{i=1}^n k_i \alpha_i$ ,  $k_i \in F$

Lemma  $k_i \in \mathbb{Q}$

pf:  $(\beta|\alpha_j) = \sum_{i=1}^n k_i (\alpha_i|\alpha_j) \quad j=1, \dots, n$

This may be viewed as a system of  $l$  equations in  $n$  unknown  $k_i$  with coefficients in  $\mathbb{Q}$ . Since the form is nondegenerate,  $\det((\alpha_i|\alpha_j)) \neq 0$ , thus the equation possess a unique solution over  $\mathbb{Q}$ .

Corollary  $\dim_{\mathbb{Q}} \sum_{\alpha \in A} \mathbb{Q}\alpha = \dim_F H^*$

Lemma  $(\cdot|\cdot) \Big|_{\sum_{\alpha \in A} \mathbb{Q}\alpha}$  is positive-definite

$$\begin{aligned} \text{pf: For any } 0 \neq \lambda \in \sum_{\alpha \in A} \mathbb{Q}\alpha, (\lambda|\lambda) &= (V^{-1}(\lambda) | V^{-1}(\lambda)) = \text{Tr}(\text{ad } V^{-1}(\lambda))^2 \\ &= \sum_{\alpha \in A} (\lambda|\alpha)^2 \geq 0 \end{aligned}$$

"=" holds  $\Leftrightarrow \lambda = 0$

Now we may extend  $\sum_{\alpha \in A} \mathbb{Q}\alpha$  to be a  $\mathbb{R}$ -vector space  $E$  by setting

$$E = \sum_{\alpha \in A} \mathbb{Q}\alpha \otimes \mathbb{R}$$

Note that the induced form also has

$$E = \sum_{\alpha \in \Delta} \alpha \otimes \alpha \in \mathbb{R}$$

Note that the induced form given by

$$(\sum v_i \otimes z_i \mid \sum w_j \otimes z_j') = \sum (v_i \mid w_j) z_i z_j'$$

is also positive-definite (the matrix is the same  $(d_i \mid d_j)$ )

$\Rightarrow E$  is a Euclidean space.

Def root system

Let  $E = \mathbb{R}^n$  be a Euclidean space. A subset  $\Delta$  is called a root system if

$$(1) \Delta \in \mathbb{R}^n \setminus \{0\}, \quad |\Delta| < \infty$$

$$(2) \text{span}_{\mathbb{R}} \{\Delta\} = \mathbb{R}^n.$$

$$(3) \forall \alpha \in \Delta, \quad S_{\alpha}(\Delta) = \Delta$$

$$(4) \forall \alpha, \beta \in \Delta, \quad \frac{2(\beta \mid \alpha)}{(\alpha \mid \alpha)} \in \mathbb{Z}$$

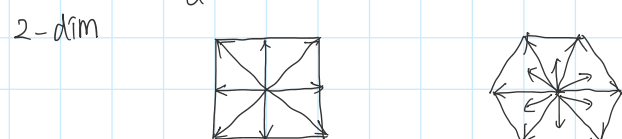
$$(5) F_{\alpha} \cap \Delta = \{\alpha, -\alpha\}$$

For now, we forget about Lie algebras.

We consider only irreducible root systems (i.e.  $\Delta \neq \Delta_1 \cup \Delta_2$  s.t.  $(\Delta_1 \mid \Delta_2) = \{0\}$ )

$$\text{e.g.} \quad \text{span}(\Delta_1) \times \text{span}(\Delta_2) = \mathbb{R}^n$$

1-dim  $\xleftarrow{-\alpha} \bullet \xrightarrow{\alpha} \Delta = \{\alpha, -\alpha\}$  the unique root system.



• angles between vectors

let  $\alpha, \beta \in \Delta$

$\theta$  = angle between them



s.t.  $(\alpha \mid \beta) \neq 0$  and  $\beta \neq \pm \alpha$

$$\text{Consider } \frac{2(\beta \mid \alpha)}{(\alpha \mid \alpha)} \in \mathbb{Z}, \quad \frac{2(\alpha \mid \beta)}{(\beta \mid \beta)} \in \mathbb{Z}$$

$$= 4 \frac{\|\beta\| \|\alpha\| \cos \theta}{\|\alpha\|^2 \|\beta\|^2} = 4 \cos^2 \theta = \begin{cases} 1 \\ 2 \\ 3 \end{cases}$$

$$\text{i.e. } \cos \theta = \pm \frac{1}{2}, \pm \frac{\sqrt{2}}{2}, \pm \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{6}, \frac{5\pi}{6}$$

In particular, at least one of  $\frac{2(\beta \mid \alpha)}{(\alpha \mid \alpha)}$ ,  $\frac{2(\alpha \mid \beta)}{(\beta \mid \beta)}$  is 1 (assume  $(\beta \mid \alpha) > 0$ )

$$\text{WLOG, assume } \frac{2(\beta \mid \alpha)}{(\alpha \mid \alpha)} = \frac{2\|\beta\|\|\alpha\|\cos \theta}{\|\alpha\|^2} = \frac{(\beta \mid \beta)}{(\alpha \mid \alpha)} = 1$$

$$\text{i.e. } \frac{\|\alpha\|}{\|\beta\|} = 2 \cos \theta \Rightarrow \cos \theta = \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}$$

$$\|\alpha\| = \|\beta\| \quad \|\alpha\| = \sqrt{2} \|\beta\| \quad \|\alpha\| = \sqrt{3} \|\beta\|$$

Def base

A subset  $B \subset \Delta$  is called a base if

$$(1) B = \{p_1, \dots, p_n\} \text{ is a basis of } \mathbb{R}^n$$

$$(2) \forall \alpha \in \Delta, \alpha = k_1 p_1 + \dots + k_n p_n \text{ where either all } k_1, \dots, k_n \in \mathbb{Z}_{\geq 0} \text{ or all } k_1, \dots, k_n \in \mathbb{Z}_{\leq 0}$$

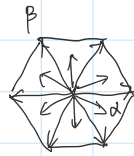
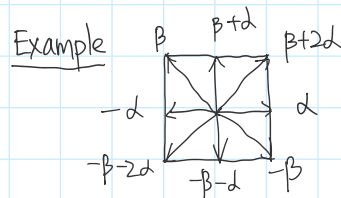
$$\Rightarrow \Delta = \Delta_+ \cup \Delta_-$$

(2)  $\forall d \in \Delta, d = k_1 \beta_1 + \dots + k_n \beta_n$  where either all  $k_1, \dots, k_n \in \mathbb{Z}_{\geq 0}$  or all  $k_1, \dots, k_n \in \mathbb{Z}_{\leq 0}$   
 $\Rightarrow \Delta = \Delta_+ \cup \Delta_-$

Lemma If  $B$  is a base,  $\beta_1 \neq \beta_2 \in B$ , then  $(\beta_1, \beta_2)$  is obtuse, i.e.  $(\beta_1 | \beta_2) \leq 0$ .

pf:  
 Suppose otherwise  $\exists \beta_1 \neq \beta_2 \in B, (\beta_1 | \beta_2) > 0$   
 Then at least one of  $\frac{2(\beta_1 | \beta_2)}{(\beta_2 | \beta_2)}, \frac{2(\beta_2 | \beta_1)}{(\beta_1 | \beta_1)}$  is 1.

WLOG, assume  $\frac{2(\beta_1 | \beta_2)}{(\beta_2 | \beta_2)} = 1$ . Then  $S_{\beta_2}(\beta_1) = \beta_1 - \frac{2(\beta_1 | \beta_2)}{(\beta_2 | \beta_2)} \beta_2 = \beta_1 - \beta_2 \in \Delta$   
 which contradicts with (2) in above def.



In general, if  $F$  is infinite,  $V = F^n$ , any finite unions of hyperplanes of  $V$  is not equal to the whole space  $V$ .

Thm existence of a base

pf:  
 First note that  $\bigcup_{d \in \Delta} d^\perp \neq \mathbb{R}^n$ .

Thus we may pick  $\gamma \in \mathbb{R}^n \setminus (\bigcup_{d \in \Delta} d^\perp)$  (such a  $\gamma$  is called regular)

i.e.  $(\gamma | d) \neq 0 \forall d \in \Delta$   
 Then  $\Delta = \Delta_+ \cup \Delta_-$ , where  $\Delta_+ = \{d \in \Delta \mid (\gamma | d) > 0\}$  and  $\Delta_- = \{d \in \Delta \mid (\gamma | d) < 0\}$ . Easy to see  $\Delta_+ = -\Delta_-$ .

Now, set  $B = \{\beta \in \Delta_+ \mid \beta \text{ is not a nontrivial sum of elements in } \Delta_+\}$

Note that if  $\beta = \beta_1 + \dots + \beta_k$  for some  $\beta_i \in \Delta_+$ , then  $(\beta | \gamma) = \sum (\beta_i | \gamma)$  where each  $(\beta_i | \gamma) > 0$ . Thus we may choose  $\beta$  s.t.  $(\beta | \gamma) = \min \{(\alpha | \gamma) \mid \alpha \in \Delta_+\} \Rightarrow \beta \in B$

Thus  $B$  is nonempty; by definition of  $B$ , it is clear that  $B$  spans  $\Delta$ . It remains to show that  $B$  is a linearly independent set. Indeed, we have the following lemma:

Lemma  $\forall s_1, s_2 \in B, (s_1 | s_2) \leq 0$ .  
 Indeed, otherwise similarly as in the proof of above lemma,  $s_1 - s_2 \in \Delta$ . If  $s_1 - s_2 \in \Delta_+$ , then  $s_1 = (s_1 - s_2) + s_2 \in \Delta_+$  contradicting with  $s_1 \in B$ ; if  $s_1 - s_2 \in \Delta_-$ , then  $s_2 - s_1 \in \Delta_+$  contradicting with  $s_2 \in B$ .

Let  $S$  be a set of nonzero vectors s.t.  $\forall s_1, s_2 \in S, (s_1 | s_2) \leq 0$  and  $\exists \gamma \neq 0, (\gamma | s) > 0$ . Then  $S$  is linearly independent.

pf:  $S = \{s_1, \dots, s_m\}$

Suppose otherwise  $\exists k_1, \dots, k_m$  not all zero s.t.  $\sum_{i=1}^m k_i s_i = 0 \Rightarrow \sum_{k_j \geq 0} k_j s_j = \sum_{k_l \leq 0} k_l s_l$

$$\begin{aligned} \text{Then } (\sum_{k_j \geq 0} k_j s_j \mid \sum_{k_l \leq 0} k_l s_l) &= (\sum_{k_j \geq 0} k_j s_j \mid -\sum_{k_l \leq 0} k_l s_l) \\ &\stackrel{V \perp 0}{=} 0 \\ &= \sum_{j \geq 0} (-k_j k_l) (s_j \mid s_l) \leq 0 \end{aligned}$$

i.e.  $(\sum_{k_j \geq 0} k_j s_j \mid \sum_{k_l \leq 0} k_l s_l) = 0 \Rightarrow \sum_{k_j \geq 0} k_j s_j = 0 \Rightarrow k_1, \dots, k_m$  can be of the same sign.

Now  $(\gamma \mid \sum_{k_i \leq 0} k_i s_i) = 0 \Rightarrow \sum_{k_i \leq 0} k_i (\gamma \mid s_i) = 0$ , a contradiction

$\alpha \in B \Rightarrow \alpha$  is a nontrivial sum of elements  $d_i$  in  $\Delta_+$ ; if  $d_i \notin B$ ,  $d_i$  is a nontrivial sum of elements in  $\Delta_+$ ; this is a finite process since  $\forall d \in \Delta, (\alpha | d) > 0$

i.e.  $(\sum k_j s_j | \sum k_j s_j) = 0 \Rightarrow \sum k_j s_j = 0 \Rightarrow k_1, \dots, k_m$  can be of the same sign.

Now  $(\gamma | \sum k_j s_j) = 0 \Rightarrow \sum_{j \geq 0} k_j (\gamma | s_j) = 0$ , a contradiction